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$$\sqrt{16} = \pm 4i$$

$$\begin{cases} x^2 - y^2 &= 0 \\ 2xy &= 4 \end{cases} \Leftrightarrow \begin{cases} x^2 - y^2 &= 0 \\ x^2(-y^2) &= -4 \end{cases}$$

$$\lambda^2 - 4 = 0$$

$$\Delta = 16$$

$$\lambda_1 = x^2 = 2$$

$$\lambda_2 = -y^2 = -2$$

$$\Rightarrow x + yi \in \{\sqrt{2} + \sqrt{2}i, -\sqrt{2} - \sqrt{2}i, \sqrt{2} - \sqrt{2}i, -\sqrt{2} + \sqrt{2}i\}$$

References